



UNIVERSIDADE ESTADUAL PAULISTA
"JÚLIO DE MESQUITA FILHO"
Campus de São José dos Campos
Instituto de Ciência e Tecnologia

Dimensionality Reduction and Feature Selection

Prof. Dr. Rogério Galante Negri

Agenda

1 High Dimensionality

2 Curse of Dimensionality

3 Dimensionality Reduction

4 Feature Selection

High Dimensionality

- Certain problems may involve data within high-dimensional feature spaces
- This circumstance can lead to the effect known as the **curse of dimensionality**
- In this context, **dimensionality reduction** and **feature selection** techniques emerge as supporting tools
- We will cover:
 - Characterization of the curse of dimensionality
 - Dimensionality reduction – PCA and LLE
 - Feature selection – suboptimal algorithms

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Curse of Dimensionality

- Increasing the number of dimensions directly affects the notion of distance
- This interference is known as the **curse of dimensionality**

Discussion

- Suppose we have a unit square, and a point is randomly selected inside it
- The probability of this point being at most 0.001 away from the square's boundary is approximately 0.4%
- Interpret the square as the feature space and the boundary region as **extreme values**
- By analogy, if we consider a unit cube, this probability increases to 0.6%

Curse of Dimensionality

Darn discussion...

- In spaces with 200, 1000, and 2000 dimensions, the respective probabilities increase to 18.1%, 63.2%, and 86.4%
- Therefore, the higher the dimensionality of the feature space, the greater the probability that a vector will present at least one extreme feature
- Considering a unit square, cube, and a 100-dimensional hypercube, the maximum distance observable between two points (along the diagonal) is $\sqrt{2}$, $\sqrt{3}$, and 10 units, respectively
- Thus, even though two points are close and contained within the same unit hypercube, they may simultaneously be distant (somewhat paradoxical)
- As a result, any given pattern tends to be far from all others

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1 High Dimensionality

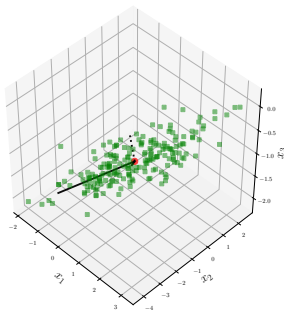
2 Curse of Dimensionality

3 Dimensionality Reduction

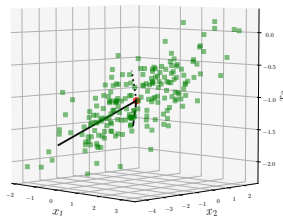
4 Feature Selection

Principal Component Analysis – PCA

- PCA performs a rotation of the axes of the original feature space
- It generates a new space defined by orthogonal axes, which are ordered based on the variance observed when projecting the dataset onto each axis



(a)



(b)

Figura: Visualization of orthogonal axes with highest variability. The solid, dashed, and dotted lines represent the axes x_1 , x_2 , and x_3 , respectively.

PCA

- Let $\mathcal{D} = \{\mathbf{x}_i \in \mathbb{R}^n : i = 1, \dots, m\}$ be a dataset with n -dimensional observations. Compute the mean vector μ and the covariance matrix Σ :

$$\mu = \frac{1}{m} \sum_{i=1}^m \mathbf{x}_i; \quad \Sigma = \frac{1}{m} \sum_{i=1}^m (\mathbf{x}_i - \mu)(\mathbf{x}_i - \mu)^T,$$

- Since Σ is a real symmetric matrix of dimension $n \times n$, it is guaranteed to have n orthogonal eigenvectors $\mathbf{v}_j$, with corresponding eigenvalues λ_j , for $j = 1, \dots, n$
- By ordering the eigenvalues in decreasing order while preserving their corresponding eigenvectors, we reindex such that $\lambda_k \geq \lambda_{k-1}$ for $k = 1, \dots, n-1$
- Based on this, we define matrix $\mathbf{W}$, where the k -th column corresponds to $\mathbf{v}_k^T$, again for $k = 1, \dots, n-1$
- $\mathbf{W}$ comprises the set of vectors responsible for rotating the original space via a linear transformation
- This reordering ensures that the first axes retain the highest levels of variability
- The projection is defined by:

$$\mathbf{x}' = \mathbf{W}^T (\mathbf{x} - \mu) \tag{1}$$

PCA

- Another important property refers to the inverse transformation, allowing $\mathbf{x}$ to be recovered from $\mathbf{x}'$
- Recalling that $\mathbf{W}$ is orthogonal, so $\mathbf{W}^{-1} = \mathbf{W}^T$, we ensure that:

$$\mathbf{x}' = \mathbf{W}^T (\mathbf{x} - \mu) \Leftrightarrow \mathbf{W}\mathbf{x}' = \mathbf{W}\mathbf{W}^T (\mathbf{x} - \mu) \Leftrightarrow \mathbf{x} = \mathbf{W}\mathbf{x}' + \mu. \quad (2)$$

- Rather than performing the inverse transformation, we may wish to retain only the first p components
- To do this, define $\mathbf{W}_p$ as the $n \times p$ matrix formed by the first p columns of $\mathbf{W}$. The projection from the original space into the new p -dimensional subspace is given by:

$$\mathbf{x}'_p = \mathbf{W}_p^T (\mathbf{x} - \mu). \quad (3)$$

PCA

An example of projection performed by the PCA method, from a three-dimensional space to a two-dimensional one.

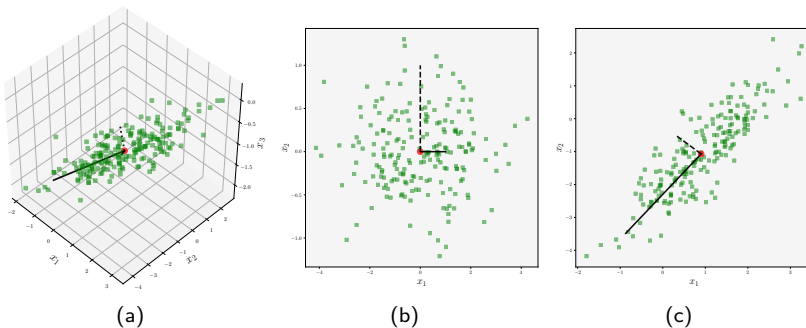


Figura: Projection from a three-dimensional space to a two-dimensional space.

Locally Linear Embedding – LLE

- A **manifold** is a topological space that locally resembles a Euclidean space.
- The LLE method assumes that the patterns lie on a smooth manifold.
- Given a dataset $\mathcal{I} = \{\mathbf{x}_i \in \mathbb{R}^n : i = 1, \dots, m\}$, the LLE method can be summarized in three main steps:
 - For each $\mathbf{x}_i$, with $i = 1, \dots, m$, define the set $\mathcal{V}_k(\mathbf{x}_i)$, consisting of the k -nearest neighbors of $\mathbf{x}_i$.
 - Then, compute a weight matrix $\mathbf{W}$, of dimension $m \times m$, aimed at reconstructing each $\mathbf{x}_i$ as a linear combination of other patterns in $\mathcal{I}$, especially those in $\mathcal{V}_k(\mathbf{x}_i)$:

$$\min_{\mathbf{W}} \sum_{i=1}^m \|\mathbf{x}_i - \sum_{j=1}^m w_{ij} \mathbf{x}_j\|^2 \quad (4)$$

subject to: $\begin{cases} w_{ij} = 0 \text{ if } \mathbf{x}_j \notin \mathcal{V}_k(\mathbf{x}_i) \\ \sum_{j=1}^m w_{ij} = 1 \end{cases}$,

LLE (cont.1)

- The elements of $\mathbf{W}$ are used to determine the projection of the patterns in $\mathcal{I}$ onto a reduced space of dimension $n^* < n$.
- The projections of each $\mathbf{x}_i$ are denoted by $\mathbf{z}_i \in \mathbb{R}^{n^*}$, for $i = 1, \dots, m$, and are determined by the matrix $\mathbf{Z}$, which solves the following problem:

$$\min_{\mathbf{Z}} \sum_{i=1}^m \left\| \mathbf{z}_i - \sum_{j=1}^m w_{ij} \mathbf{z}_j \right\|^2 \quad (5)$$

LLE

A typical example of applying the LLE method, where a nonlinear surface is remapped onto a plane. Neighborhood relationships between patterns are indicated by their associated colors and are preserved after projection.

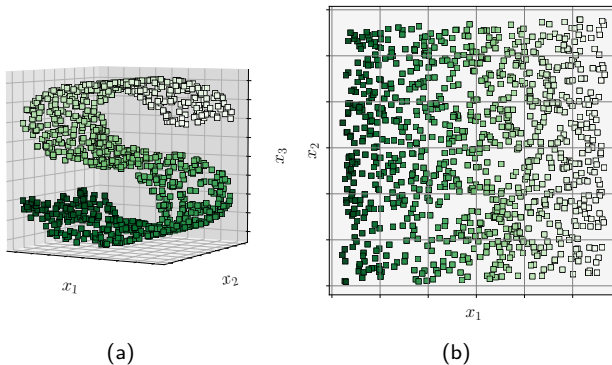


Figura: Example of remapping performed by the LLE method.

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Feature Selection

- When the feature space has high dimensionality, the effects of the curse of dimensionality and computational resource limitations naturally raise the question: **what is the most appropriate subset of features for solving the given classification problem?**
- When selecting a fixed subset of s features from a total of n , the number of possible combinations is $\binom{n}{s} = \frac{n!}{s!(n-s)!}$, which, depending on n and s , may be prohibitively large and computationally infeasible
- In summary, the process becomes impractical, thus motivating the development of strategies that explore the space of possibilities toward the best feature subset at an acceptable computational cost

Feature Selection

- Feature selection methods are designed to address this type of problem and are typically categorized as:
 - **Optimal** – involving exhaustive search and optimization strategies such as *branch-and-bound*, *simulated annealing*, and genetic algorithms
 - **Sub-optimal** – which seek non-optimal solutions in favor of computational efficiency
- The process is also guided by a cost function, which relates a given subset of features to its corresponding class separability capability

Scatter Matrix-Based Measures

- Intra- and inter-class scatter matrices have previously been used (in clustering and evaluation contexts)
- These matrices are useful for quantifying class separability
- Adapted to the context of feature selection and assuming a labeled dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i) : i = 1, \dots, m\}$, we define:

$$\mathbf{V}_I = \frac{1}{m} \sum_{j=1}^c \sum_{i=1}^m \delta(\mathbf{x}_i, \omega_j) (\mathbf{x}_i - \mu_j)^T (\mathbf{x}_i - \mu_j), \quad (6)$$

$$\mathbf{V}_E = \sum_{j=1}^c \frac{m_j}{m} (\mu - \mu_j)^T (\mu - \mu_j), \quad (7)$$

- It is reasonable to define measures that favor higher class separability by minimizing intra-class scatter and maximizing inter-class scatter
- Measures aligned with this rationale include:

$$J = \text{Tr}(\mathbf{V}_I^{-1} \mathbf{V}_E) \quad J = \frac{|\mathbf{V}_E + \mathbf{V}_I|}{|\mathbf{V}_I|} \quad J = \frac{\text{Tr}(\mathbf{V}_E)}{\text{Tr}(\mathbf{V}_I)} \quad (8)$$

Sub-Optimal Feature Selection Approaches

- Given a feature space $\mathcal{X}$ of dimension n , patterns in this space are vectors of the form $\mathbf{x} = [x_1, \dots, x_n]$, where each component x_i is associated with a feature ξ_i
- We can define $\Xi = \{\xi_1, \dots, \xi_n\}$ as the set of features that characterize $\mathcal{X}$
- Considering a set of labeled examples $\mathcal{D} = \{(\mathbf{x}_i, y_i) \in \mathcal{X} \times \mathcal{Y} : i = 1, \dots, m\}$, and assuming the feature selection process reduces the dimensionality of $\mathcal{X}$, the patterns $\mathbf{x}_i$ will have fewer components, but the pairings $(\mathbf{x}_i, y_i)$ remain valid
- This allows us to evaluate class separability based on a subset $\hat{\Xi} \subseteq \Xi$ of features

Sequential Forward/Backward Selection

Sequential Forward Selection – SFS

- The SFS algorithm performs a *bottom-up* search, selecting one feature at a time from the n available features until s features are selected
- The selection is guided by maximizing class separability, measured by a cost function J
- Thus, given a feature subset $\hat{\Xi}$ initially empty, features $\xi_\ell \in \Xi$ are selected and added to $\hat{\Xi}$ as long as:

$$\arg \max_{\substack{\ell=1,\dots,n \\ \xi_\ell \notin \hat{\Xi}}} J[\hat{\Xi} \cup \xi_\ell]. \quad (9)$$

Sequential Forward/Backward Selection

Sequential Backward Selection – SBS

- A *top-down* search that begins with $\hat{\Xi} = \{\xi_1, \dots, \xi_n\}$ and eliminates features one by one until only s remain
- The elimination process aims to minimize the loss of class separability; that is, ξ_ℓ is removed from $\hat{\Xi}$ if:

$$\arg \max_{\substack{\ell=1, \dots, n \\ \xi_\ell \notin \hat{\Xi}}} J[\hat{\Xi} \setminus \xi_\ell]. \quad (10)$$

SFS vs. SBS

SFS Algorithm Steps:

Given $\mathcal{D}$ and s

$$\Xi = \{\xi_1, \dots, \xi_n\}$$

$$\hat{\Xi} = \emptyset$$

While $\#\hat{\Xi} < s$:

$$\xi_\ell = \arg \max_{\substack{\ell=1, \dots, n \\ \xi_\ell \notin \hat{\Xi}}} J[\hat{\Xi} \cup \xi_\ell]$$

$$\Xi := \Xi \setminus \{\xi_\ell\}$$

$$\hat{\Xi} := \hat{\Xi} \cup \{\xi_\ell\}$$

SBS Algorithm Steps:

Given $\mathcal{D}$ and s

$$\Xi = \{\xi_1, \dots, \xi_n\}$$

$$\hat{\Xi} := \Xi; \Xi := \emptyset$$

While $\#\hat{\Xi} > s$:

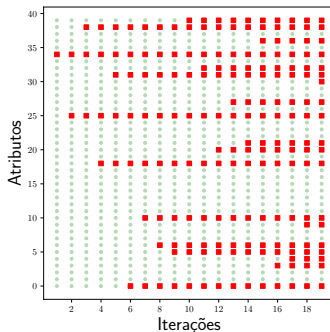
$$\xi_\ell = \arg \max_{\substack{\ell=1, \dots, n \\ \xi_\ell \in \hat{\Xi}}} J[\hat{\Xi} \setminus \xi_\ell]$$

$$\Xi := \Xi \cup \{\xi_\ell\}$$

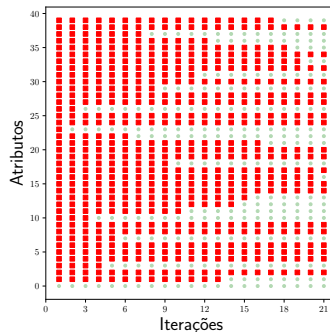
$$\hat{\Xi} := \hat{\Xi} \setminus \{\xi_\ell\}$$

SFS vs. SBS

Selection of 19 features from a simulated dataset with 40 features.



(a) SFS



(b) SBS

Figura: Feature subset selection using the SFS and SBS algorithms. Dark squares and light circles respectively denote selected and non-selected features.

Floating Processes

- To overcome the limitation of not being able to “undo” a selection or elimination, the SFFS and SBFS (floating) variants are proposed
- In SFFS, assuming a feature subset $\hat{\Xi}$ initially empty, a feature $\xi_{\ell_1} \notin \hat{\Xi}$ is selected according to the **inclusion criterion** and added to $\hat{\Xi}$
- Next, the feature $\xi_{\ell_2} \in \hat{\Xi}$ is identified, whose removal, according to the **exclusion criterion**, results in the least degradation of class separability
- If $\xi_{\ell_1} = \xi_{\ell_2}$, removal is not performed and a new feature is considered for inclusion
- On the other hand, if $\xi_{\ell_1} \neq \xi_{\ell_2}$ and $\#\hat{\Xi} > 1$, then ξ_{ℓ_2} is removed, and further removal attempts for other features $\xi_{\ell} \in \hat{\Xi}$ continue as long as separability improves, i.e., $J[\hat{\Xi} \setminus \xi_{\ell}] > J[\hat{\Xi}]$ or $\#\hat{\Xi} = 1$

SFFS in Action

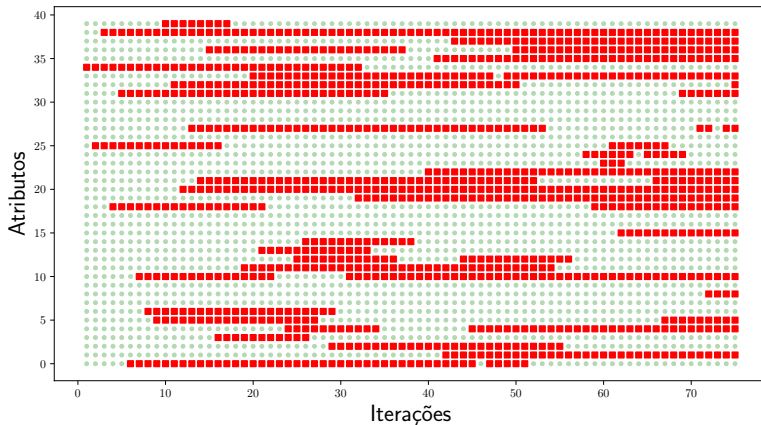


Figura: Feature subset selection using the SFFS algorithm. Dark squares and light circles respectively denote selected and non-selected features.

Dimensionality Reduction and Feature Selection

rogerio.negri@unesp.br